Activity-Target Dependence in the Structural Selection of Two-Dimensional ϕ^4 Field Configurations

Dimensional Dependence and Selection Transition
in the Activity Sector

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Abstract

We extend the structural selection framework of the companion papers [1, 2] to two spatial dimensions, revealing a clear selection transition in the activity target parameter $s_\star$. In a two-dimensional ϕ^4 scalar field theory on a 48×48 lattice, four classes of field configurations are evolved under conservative Klein–Gordon dynamics: homogeneous vacuum states, domain-wall configurations, localised wave packets, and chaotic initial data.

A systematic sweep over the activity target parameter $s_\star \in \{0.01, 0.02, 0.05, 0.10, 0.20, 0.42\}$ reveals a clear selection transition: for $s_\star \lesssim 0.20$, domain-wall configurations are the dominant class in the reference summaries and in the majority of tested (λ_i, β) settings, capturing 72%–87% of the total Boltzmann weight; for $s_\star = 0.42$, chaotic configurations dominate (100% weight).

This result establishes that the structural selection principle possesses a dimensional dependence: the activity target $s_\star$ that favours topologically structured configurations in one dimension must be recalibrated for two dimensions, where domain walls are significantly less kinetically active. We interpret this as a *structural selection transition* in the activity sector, analogous to thermodynamic phase transitions driven by temperature.

The study is presented as a proof of concept; the determination of $s_\star^c$ from first principles remains an open question.

1 Introduction

The structural selection framework proposed in [1] assigns a Boltzmann-type probability weight to field configurations based on a structural energy functional $\mathcal{E}[\Phi]$:

$$P[\Phi] \propto \exp(-\beta\, \mathcal{E}[\Phi]). \tag{1}$$

In the one-dimensional companion study [2], kink configurations dominated the structural weight across all 81 tested parameter combinations, providing a robust proof of concept for the framework's discriminative power.

A natural question arises: does this result persist in higher dimensions? In two spatial dimensions, the topological analogue of a kink is a *domain wall* — a planar interface separating two vacuum regions. Domain walls share the topological protection of kinks but differ in their dynamical properties: in 2D, the field energy distributes over N^2 lattice sites rather than N, leading to significantly smaller mean kinetic energy per degree of freedom.

This dimensional difference has a direct consequence for the structural selection: the activity target $s_\star$ that correctly identifies kinks as preferred in 1D may not correctly identify domain walls as preferred in 2D, since domain walls are effectively less active in mean-normalised units.

The present paper investigates this question systematically through a sweep over $s_\star$, revealing a *structural selection transition* in the activity sector.

2 Model and Setup

2.1 Field Theory

We study the ϕ^4 scalar field on a periodic 48×48 lattice with spacing $\Delta x = 1$:

$$\ddot{\phi}_{ij} = \frac{\phi_{i+1,j} + \phi_{i-1,j} + \phi_{i,j+1} + \phi_{i,j-1} - 4\phi_{ij}}{\Delta x^2} - V'(\phi_{ij}), \quad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2. \tag{2}$$

Dynamics is integrated with a fourth-order Runge–Kutta scheme ($\Delta t = 0.02$, $T_{\max} = 16$).

2.2 Configuration Classes

Four classes of initial conditions are studied:

Vacuum: $\phi \approx \pm 1$ with small noise.

Domain wall: $\phi(x, y) = \tanh((x - x_0)/w)$ interpolating between $\phi = -1$ and $\phi = +1$, with width $w \in [4, 10]$ and centre $x_0 \in [-10, 10]$.

Localised: Gaussian wave packet, $\phi \propto \exp(-r^2/(2\sigma^2))$, non-topological and dispersive.

Chaotic: Random initial data, $\phi, \dot{\phi} \sim \mathcal{N}(0, 1)$.

For each class, $n = 12$ independent runs are generated.

2.3 Structural Diagnostics

The structural energy uses three lattice-adapted sectors:

$$\mathcal{E}_{\text{stab}} = \frac{\sum_{ij} \pi_{ij}^2}{\Lambda_{\text{stab}} + \sum_{ij} \pi_{ij}^2} + \frac{\sum_{ij} b_{ij}^2}{\Lambda_{\text{bound}} + \sum_{ij} b_{ij}^2}, \tag{3}$$

$$\mathcal{E}_{\text{conn}} = -\tfrac{1}{2}\left[I(\phi_{i,j} : \phi_{i+1,j}) + I(\phi_{i,j} : \phi_{i,j+1})\right], \tag{4}$$

$$\mathcal{E}_{\text{dyn}} = \left(\frac{s_{\text{dyn}} - s_\star}{\sigma_s}\right)^2, \tag{5}$$

where $b_{ij} = \max(0, |\phi_{ij}| - \phi_{\max})$, I denotes mutual information between adjacent field values, $s_{\text{dyn}} = \langle\dot{\phi}^2\rangle/(10 + \langle\dot{\phi}^2\rangle + \alpha\langle|\nabla\phi|^2\rangle)$ is a mean-based activity score, and $s_\star$ is the *activity target parameter* whose role is the subject of this paper. The total structural energy is

$$\mathcal{E} = \lambda_{\text{stab}}\,\mathcal{E}_{\text{stab}} + \lambda_{\text{conn}}\,\mathcal{E}_{\text{conn}} + \lambda_{\text{dyn}}\,\mathcal{E}_{\text{dyn}}. \tag{6}$$

3 Activity Target Sweep

3.1 Measured Activity Levels

Before performing the sweep, we measure the mean dynamical activity $\bar{s}_{\text{dyn}}$ for each configuration class at the reference parameter setting $\lambda_i = 1$, $\beta = 4$:

Table 1: Mean activity scores in 2D ($n = 12$ runs per class).

Class	$\bar{s}_{\text{dyn}}$	Distance from $s_\star = 0.42$
Vacuum	0.0002	0.420
Domain wall	0.0071	0.413
Localised	0.0175	0.403
Chaotic	0.2297	0.190

All structured configurations (vacuum, domain wall, localised) are far less active than the target $s_\star = 0.42$, while chaotic configurations are closest to the target. This immediately explains why $s_\star = 0.42$ fails to select domain walls in 2D.

3.2 Phase Diagram

Table 2 reports the class weights for each value of $s_\star$ at the reference parameter setting.

Table 2: Boltzmann weight per class as a function of activity target $s_\star$ ($\lambda_i = 1$, $\beta = 4$). Domain wall dominates for $s_\star \leq 0.20$; chaotic dominates for $s_\star = 0.42$.

$s_\star$	Vacuum	Domain wall	Localised	Chaotic
0.01	0.042	**0.871**	0.087	≈ 0
0.02	0.042	**0.870**	0.089	≈ 0
0.05	0.039	**0.865**	0.095	≈ 0
0.10	0.036	**0.857**	0.107	≈ 0
0.20	0.025	**0.725**	0.117	0.133
0.42	≈ 0	≈ 0	≈ 0	**1.000**

The transition is sharp: domain walls dominate consistently for $s_\star \in [0.01, 0.20]$, capturing 72%–87% of the total structural weight. At $s_\star = 0.42$, the chaotic class captures all weight. The transition region $s_\star^c$ lies in the interval $(0.20, 0.42)$.

3.3 Robustness

To test whether the phase diagram is robust across parameter choices, we perform a sweep over $\lambda_{\text{stab}}, \lambda_{\text{conn}}, \lambda_{\text{dyn}} \in \{0.5, 1.0, 2.0\}$ and $\beta \in \{2.0, 4.0, 8.0\}$ for each value of $s_\star$. This yields $81 \times 6 = 486$ total parameter combinations.

The result confirms the phase diagram: domain walls are the dominant class in the majority of parameter settings for $s_\star \leq 0.20$, while chaotic configurations dominate for $s_\star = 0.42$.

4 Discussion

4.1 Dimensional Dependence of the Activity Target

The key finding is that the activity target $s_\star$ is *not a universal constant* of the structural selection framework, but a *dimension-dependent calibration parameter*.

In one dimension [2], kink configurations have sufficient mean kinetic activity to be selected by $s_\star = 0.42$. In two dimensions, domain walls have much lower mean kinetic energy per degree of freedom:

$$\bar{s}_{\text{dyn}}^{\text{1D,kink}} \approx 0.255, \qquad \bar{s}_{\text{dyn}}^{\text{2D,wall}} \approx 0.007. \tag{7}$$

This factor of ~ 35 reduction in activity reflects the redistribution of kinetic energy over N^2 rather than N degrees of freedom.

4.2 Structural Selection Transition

The sharp transition between domain-wall dominance and chaotic dominance as $s_\star$ increases from 0.20 to 0.42 constitutes a *structural selection transition* in the activity sector.

By analogy with thermodynamic phase transitions (noting that this is an analogy in the selection parameter space, not a thermodynamic transition in the strict sense):

- $s_\star$ plays the role of *temperature*: low $s_\star$ favours ordered, low-activity structures; high $s_\star$ favours active, disordered configurations.
- The transition region $s_\star \in (0.20, 0.42)$ separates two phases of the structural selection landscape.

- Below $s_\star^c$: topological order is preferred. Above $s_\star^c$: dynamical activity is preferred.

This analogy suggests that the structural selection framework may possess a rich selection-transition structure as a function of its parameters, analogous to statistical mechanical systems with order-disorder transitions.

4.3 Implications for the General Framework

The dimensional dependence of $s_\star$ raises an important question for the general framework: how should the activity target be determined from first principles?

Two natural approaches suggest themselves:

1. **Dimensional calibration:** $s_\star$ is set by the typical kinetic activity of the topological ground state of the theory in d dimensions. This would make $s_\star$ theory-dependent and dimension-dependent, but derivable from the dynamics.

2. **Information-balance condition:** $s_\star$ is determined by a self-consistency condition on the MaxEnt distribution, requiring that the selected configurations saturate a prescribed information-production rate. This would make $s_\star$ an emergent parameter rather than a free input.

Both approaches are deferred to future work.

4.4 Limitations

The present study has the following limitations:

1. **Single model.** The selection transition is demonstrated for a ϕ^4 field on a 48×48 lattice. Extension to other potentials and lattice sizes is required.

2. **Approximate $s_\star^c$.** The transition region is bracketed as $s_\star \in (0.20, 0.42)$ but not precisely determined. A finer sweep would localise the transition more precisely.

3. **Benchmark parameters.** The activity width $\sigma_s = 0.18$ and the Lagrange multipliers λ_i are benchmark choices; the phase boundary may shift under different parameter settings.

5 Conclusion

We have demonstrated that the structural selection framework of [1] exhibits a *structural selection transition* in two spatial dimensions as a function of the activity target $s_\star$.

For $s_\star \lesssim 0.20$, domain-wall configurations are the dominant class throughout the tested low-$s_\star$ benchmark region, consistent with their topological stability and spatial coherence. For $s_\star = 0.42$, chaotic configurations dominate, reflecting their higher mean kinetic activity.

The transition reveals that the activity target $s_\star$ is a dimension-dependent calibration parameter: it must be set to match the typical activity level of the topological ground state in the given dimension, rather than treated as a universal constant.

This finding enriches the structural selection framework with a phase structure and raises the question of whether $s_\star$ can be derived from first principles — either through dimensional calibration or through an information-balance self-consistency condition.

Both directions are left as open problems for future work.

The 2D study shows more broadly that the discriminative behaviour of the structural selection functional depends sensitively on the calibration of the activity sector. In particular, the preferred activity target shifts strongly with spatial dimension — a finding that motivates a principled derivation of $s_\star$ from the underlying field theory rather than treating it as a free parameter.

A Reproducibility

All results are fully reproducible from the companion code:

https://github.com/Chris4081/structural-selection-principle

Key parameters:

$$
\begin{aligned}
N = 48, \quad \Delta x = 1, \quad \Delta t = 0.02, \quad T_{\max} = 16, \\
n = 12 \text{ runs per class}, \quad \text{seed} = 42, \\
s_\star \in \{0.01, 0.02, 0.05, 0.10, 0.20, 0.42\}, \quad \sigma_s = 0.18, \\
\phi_{\max} = 3, \quad \Lambda_{\text{stab}} = \Lambda_{\text{bound}} = 10.
\end{aligned}
$$

Run with:

```
pip install numpy pandas matplotlib
python maat_structural_selection_2d_sstar_sweep.py
```

References

[1] C. Krieg, *A Structural Selection Principle for Dynamically Consistent Field Configurations*, preprint, available from the author (2026).

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